DO NOW — RECALL

Lesson 070 · Heuristics and Performance Modelling

Name: _____ Class: _____ Date: _____

1. What is correlation?

2. What does a heuristic provide?

3. What is an optimisation objective?

4. A sorted 32-element candidate set is halved to one candidate. How many halvings are needed?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Statistical association, not proof of causation.

2. Model answer

An estimate guiding an algorithm's next choice.

3. Model answer

A quantity to minimise/maximise.

4. Model answer

Five: 32→16→8→4→2→1.

By the end, you can…

• A heuristic is a useful estimate or rule of thumb that guides decisions when complete exploration is expensive.

• A locally cheapest edge does not necessarily lead to a globally cheapest route; evaluate candidate routes as complete paths.

• Performance modelling estimates time, memory or cost under assumptions. A predicted result requires checking against real hardware and workloads.

Vocabulary — know these terms

heuristic — Estimate guiding solution search.

greedy — Choose apparently best immediate option.

performance model — Simplified resource/time estimate.

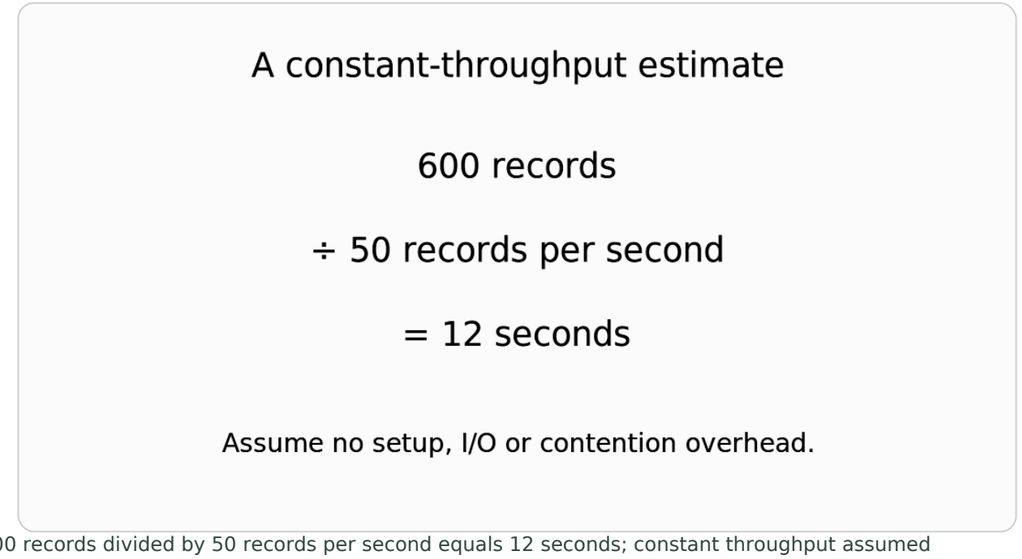
admissible — Heuristic never overestimates remaining optimal cost.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Estimate processing 600 records at 50 records/s, then discuss what model ignores.

A performance model is a simplified prediction, not a guaranteed elapsed time.



600 records divided by 50 records per second equals 12 seconds; constant throughput assumed

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

Follow the reasoning
A performance model is a simplified prediction, not a guaranteed elapsed time.

NEW LEARNING

Read, annotate and explain

Define and predict

A heuristic is a useful estimate or rule of thumb that guides decisions when complete exploration is expensive.

Explain

A locally cheapest edge does not necessarily lead to a globally cheapest route; evaluate candidate routes as complete paths.

Evaluate

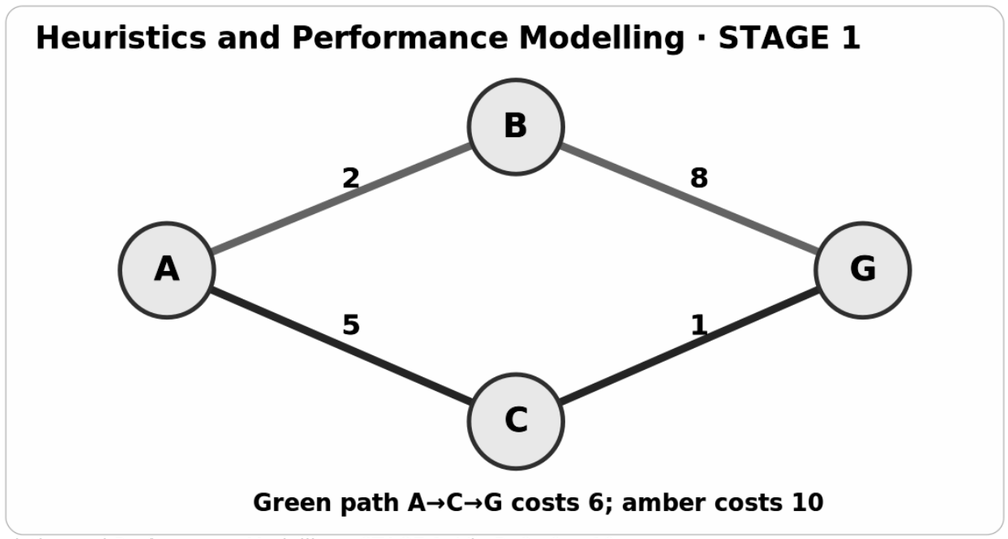
Performance modelling estimates time, memory or cost under assumptions. A predicted result requires checking against real hardware and workloads.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Heuristics and Performance Modelling · STAGE 1; Via B: 2+8 = 10

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A local greedy choice always produces an optimal complete path.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

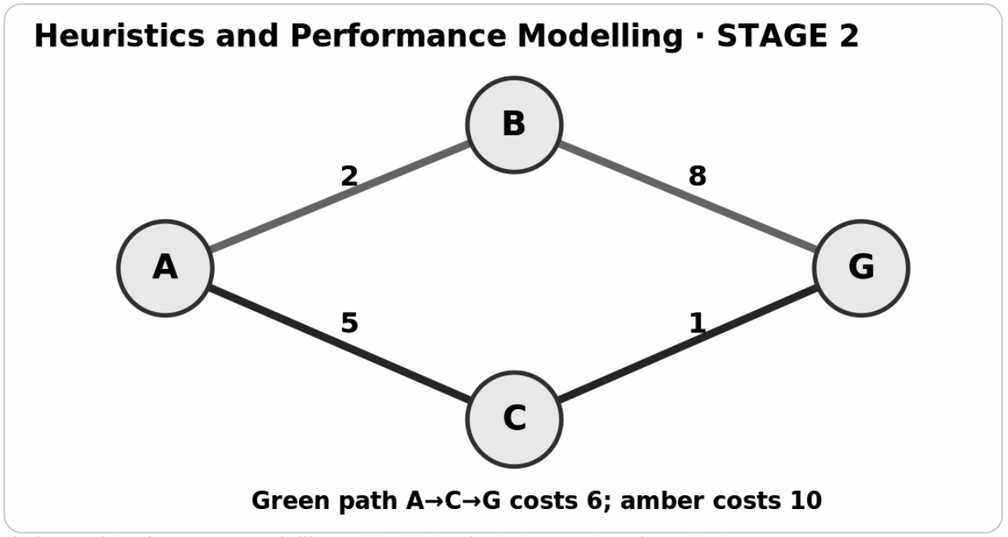
Question 3

Evaluate using straight-line distance as heuristic for car routes with one-way roads and congestion. [6]

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning

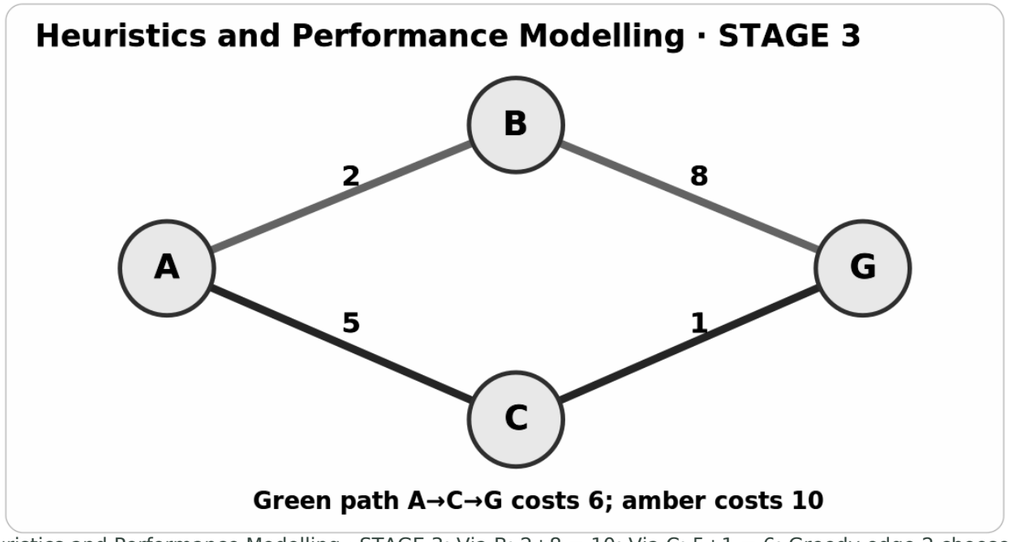


Heuristics and Performance Modelling · STAGE 2; Via B: 2+8 = 10; Via C: 5+1 = 6

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Heuristics and Performance Modelling · STAGE 3; Via B: 2+8 = 10; Via C: 5+1 = 6; Greedy edge 2 chooses B first

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Compare route A-B-G edges 2+8 against A-C-G 5+1. Calculate totals and select cheaper. [6]

Question 2

Model 1200 jobs processed at 80/s, under constant throughput: estimate duration and one assumption. [6]

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

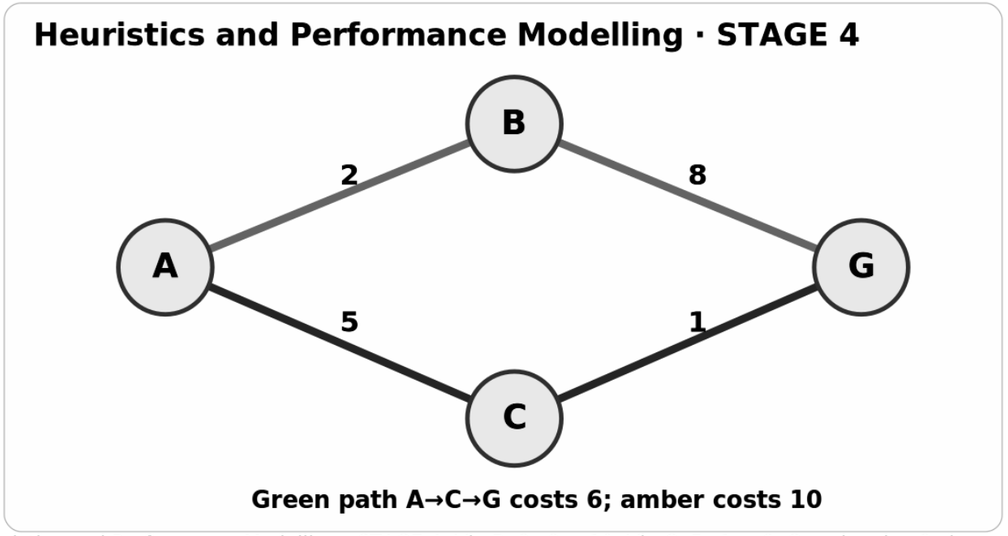
Guided 1 — TRY

Estimate processing 600 records at 50 records/s, then discuss what model ignores.
Source code / input:
600/50=12s under constant-rate assumptions; ignores setup/IO/congestion.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Heuristics and Performance Modelling · STAGE 4; Via B: 2+8 = 10; Via C: 5+1 = 6; Greedy edge 2 chooses B first; Full route via C is cheaper

I DO — WORKED EXAMPLE

Follow the worked method

Choose cheaper complete path: X→Y→G is 3+9; X→Z→G is 6+2.

Follow the reasoning

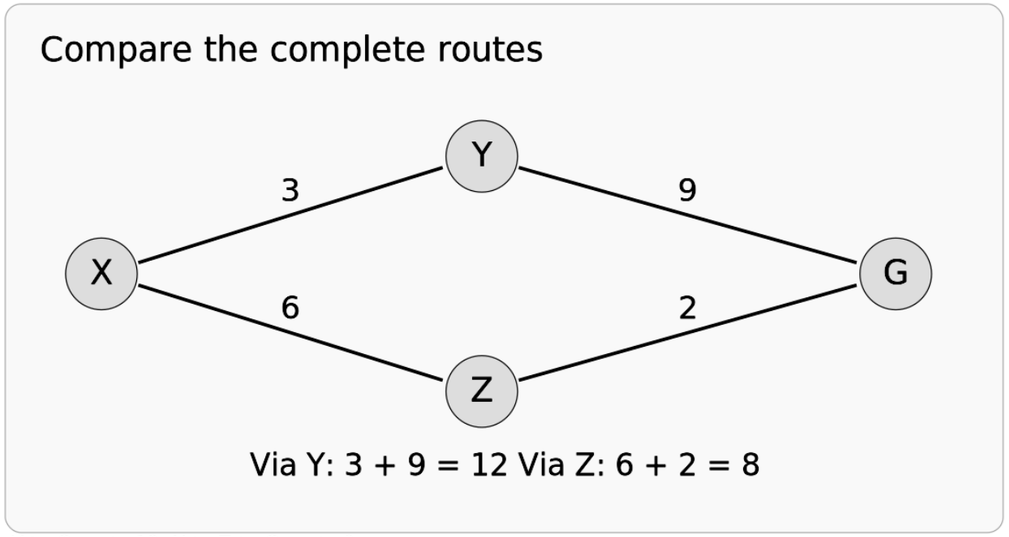
Calculate the entire feasible route rather than the next immediate edge.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose cheaper complete path: X→Y→G is 3+9; X→Z→G is 6+2.

Calculate the entire feasible route rather than the next immediate edge.



X to Y to G costs 12; X to Z to G costs 8